

Aafthab Ali Shaik

Software Engineer

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🌐 LinkedIn

🐙 GitHub

🌐 Portfolio

PROFESSIONAL SUMMARY

I am a Computer Science Student and I have experience in building applications that can handle a lot of users and machine learning systems. I know a lot about the basics of Computer Science, like data structures and algorithms. I also know how to build the backend of a website make APIs and design databases. I have built applications that're ready to be used by people using React.js, Node.js and TensorFlow. I want to be a Software Engineering Intern to help make systems that work well and can handle a lot of users.

TECHNICAL SKILLS

Programming: C, Python, Java, C Sharp

Frontend: React.js, TypeScript, JavaScript, Vite

Backend: Node.js, Express.js, Flask, FastAPI, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, Firebase, Supabase

Cloud & Tools: Git, GitHub, GitLab CI/CD, GitLab Runner, Docker, Render, Vercel, Hugging Face Transformers.js

Testing: Pytest, Playwright, Vitest, React Testing Library

Core Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Static Code Analysis, Embedded Software Build Automation

EXPERIENCE

• **Defence Research and Development Organisation (DRDO) Research Centre Imarat (RCI)** *May 2026 – Jul 2026*
Software Engineering Intern Hyderabad

- Designed and developed a CI/CD pipeline for embedded software build, verification, and static analysis using GitLab CI/CD.
- Configured GitLab Runners and Docker containers to automate embedded software build workflows.
- Performed static code analysis and generated automated build reports to improve software quality.
- Collaborated with DRDO scientists on embedded software development and DevOps workflow optimization.

• **Centre for Extended Reality (CXR), GITAM University** *May 2026 – Jun 2026*
Full Stack Developer Intern Visakhapatnam

- Developed and enhanced the **CXR Web Portfolio** by improving both frontend and backend user interfaces for a responsive and user-friendly experience.
- Integrated Firebase services by implementing API calls for data retrieval, storage, authentication, and real-time updates while ensuring seamless frontend-backend communication.
- Refactored the codebase by fixing bugs, improving code quality, resolving security vulnerabilities, and optimizing application performance for a stable production-ready deployment.

• **Suvehi Interactive Technologies Pvt. Ltd.** *Jan 2026 – Apr 2026*
Software Engineering Intern Visakhapatnam

- Designed and developed an internal company website by implementing responsive frontend interfaces and scalable backend functionalities.
- Worked on multiple software development projects, including website development, game version upgrades, feature implementation, software maintenance, and bug fixing.

• **Infosys Springboard** *Dec 2025 – Feb 2026*
Full Stack Developer Intern

- Designed and implemented a full-stack web application using React.js, Node.js, and MySQL.
- Developed RESTful APIs and optimized backend request handling.
- Implemented JWT-based authentication and integrated Chart.js for financial data visualization in TaxPal project.

• **BharatVersity** *Apr 2025 – Jul 2025*
AI/ML Intern BITS Hyderabad

- Built a Music Recommendation System using Retrieval-Augmented Generation (RAG).
- Implemented data preprocessing, feature engineering, and model evaluation pipelines.
- Automated ML workflows and structured data pipelines for improved retrieval accuracy.

PROJECTS

- AI Voice Assistant**

Dec 2025

Developed an AI-powered voice assistant integrating OpenAI and Gemini APIs with real-time speech processing and response generation.
- TaxPal Personal Finance & Tax Estimator**

Feb 2026

Developed a full-stack web application using React.js, Node.js, and MySQL with RESTful APIs for managing income, expenses, and tax estimation. Implemented JWT-based authentication for secure user access and integrated Chart.js for visualizing financial insights through interactive dashboards.
- Real-Time Facial Expression Detection Web App**

Sep 2026

Built and deployed a real-time facial expression detection web application using React.js, Flask, ONNX Runtime, and MediaPipe. Implemented live camera-based emotion detection, Android/iOS browser compatibility fixes, GitHub Actions testing, Docker deployment, and Render keep-alive health checks.

–Live Project Link

–GitHub Repository Link
- RouteWise – Smart Journey Planner with Learning Fares**

Sep 2026

Built and deployed a real-time journey planner for Visakhapatnam using React, TypeScript, FastAPI, Google Maps APIs, and Supabase PostgreSQL. Implemented multi-modal route comparison for buses, autos, cabs, bikes, and walking based on estimated cost and travel time, along with a Bayesian fare model using rider fare votes, outlier rejection, and time-decayed weighting. Added live GPS navigation with off-route detection, Google/Apple Maps hand-off, Docker deployment on Render, and 62 automated tests.

–Live Project Link

–GitHub Repository Link
- StudyMind – AI Study Workspace & Interview Coach**

Sep 2026

Built and deployed an AI study platform using React.js, Vite, Firebase, and the Google Gemini API. Implemented PDF parsing with in-browser OCR (Tesseract.js) and hybrid BM25 + MiniLM retrieval for page-cited answers, generating notes, 2/5/8-mark question banks, and document-grounded quizzes. Built an Interview Studio with adaptive Gemini follow-ups and on-device speech recognition (Whisper via Transformers.js). Added an admin dashboard, Firestore security rules, 10x faster uploads, 160 Vitest tests, and Render deployment.

–Live Project Link

–GitHub Repository Link

EDUCATION

- GITAM University**

Aug 2023 – Present

B.Tech in Computer Science / CGPA: 8.06

Visakhapatnam

CERTIFICATIONS

- Microsoft Azure AI Fundamentals (AI-900)**

Apr 2024

Score: 781

LEADERSHIP

- Core Member Google Developers Group (GDG), GITAM University

2025 – Present